Gruhesh Sri Sai Karthik Kurra

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Imperial MSc (AI and ML) student working on language models, available from autumn 2027. First author of an NLP paper in IEEE Xplore and of two papers accepted at IEEE NEPCON 2026, on diverse generation from a frozen transformer and on deepfake detection. Ran a data-leakage audit that changed one paper's main result. Looking for research engineer roles in LLMs.

Education

Imperial College London, United Kingdom Sep 2026 - Sep 2027 (expected)
MSc Computing (Artificial Intelligence and Machine Learning), Department of Computing | Enrolled

KL University, Hyderabad, India Aug 2021 - May 2025
B.Tech Computer Science and Engineering | CGPA 9.72 / 10

iHub-Data, IIIT Hyderabad, India May 2024 - Oct 2024
Six-month machine learning programme: classifiers, text representation, optimisation, clustering, PCA, probabilistic models, deep learning

Selected Publications

Morphology-Aware Nested Character Embeddings for Word Representation (MANCE)
Gruhesh Sri Sai Karthik Kurra, Chandanasree Moparthi, Prathipati Manish Chowdary, Pavan Kumar Pagadala
IEEE ICCCMLA 2025, published in IEEE Xplore
ieeexplore.ieee.org/document/11580466 | Code: github.com/GruheshKurra/MANCE-NLP

- Represented each word as an ordered sequence of character embeddings, so morphological variants share part of their representation. The vocabulary is about 100 characters.
- Re-implemented it as an open, seeded PyTorch pipeline (character Transformer word encoder, 3 seeds). It scored 63.0 F1 on GermEval 2014 German NER against 48.2 for word embeddings with the same encoder, and was on par on AG News (91.0%) and DBpedia (98.5%).
- Ran ablations on AG News, where averaging characters instead of encoding them in order cost 5.6 points.

Prism Tuning: Repulsion-Trained Seed Embeddings in a Frozen Transformer for Non-Redundant Generation
Gruhesh Sri Sai Karthik Kurra, Pavan Kumar Pagadala, Malathy Batumalay, Sushma Reddy Koduru
Accepted, IEEE NEPCON 2026 (presentation December 2026)

- Proposed adding K seed vectors to the input embeddings of a frozen transformer and training only the seeds, with a repulsion loss. One prompt then gives K different outputs in one batched pass, with no filtering at inference.
- Made the loss trainable through a hidden-state surrogate at generated positions. Whitened representations to counter anisotropy, and split each seed into shared style and repelled content parts. The paper covers architecture and formulation only.
- Designed and ran a follow-up, Prism-MCL (Oct 2026). It replaced the loss with multiple-choice learning and tested prefix and per-layer attention seeds with LoRA on Qwen2.5 0.5B and 3B on RunPod. Pre-registered runs on 42 held-out categories found no significant gain over tuned min-p sampling.

Dual-Stream Artifact Detection with Iterative Evidence Refinement for Frame-Level Deepfake Recognition
Gruhesh Sri Sai Karthik Kurra, Pavan Kumar Pagadala, Malathy Batumalay, Veda Boddapati
Accepted, IEEE NEPCON 2026 (presentation December 2026)
Frame corpus: kaggle.com/datasets/gruheshkurra/radar-deepfake-frames

- Audited the data split and found that 97.3% of WildDeepfake test frames shared source videos with training. Reported the video-disjoint FaceForensics++ and Celeb-DF test subset (5,680 frames) as the main result, with 96.3% AUC, and 87.1% on Celeb-DF.
- Built a ViT-Small detector with Sobel-edge and three-band FFT evidence streams, an orthogonality penalty and iterative cross-attention fusion. Trained it on 227,504 frames from FaceForensics++, Celeb-DF v2 and WildDeepfake.

BIE: Bit-Index Encoding of Sparse Neural Network Weights

Gruhesh Sri Sai Karthik Kurra

Preprint, Zenodo, version 2, October 2026

doi.org/10.5281/zenodo.17217217 | Code: github.com/GruheshKurra/bit-index-encoding-research-

- Designed a storage format for pruned weights, with Golomb–Rice coded positions, 8-bit codes stored as bitplanes and a row-parallel kernel. It is $44.5\times$ smaller than 32-bit weights at 95% sparsity with 0.6% error, against $25.7\times$ for CSR at the same error. Perplexity of a pretrained 124M-parameter language model stayed at the int8 level (30.0 vs 29.9).

Research Experience

Research Intern | IIIT Hyderabad Dec 2024 - Jun 2025

Hyderabad, India

- Designed a document translation system that keeps the original layout, using Vision Transformer layout detection, character-metric font detection and K-means colour extraction.
- Built the optical character recognition pipeline with FastAPI and Tesseract for English, Hindi, Telugu and German. The output keeps source fonts, colours and page structure.
- Demonstrated the system live to faculty and industry visitors at the IIIT Hyderabad research expo, May 2025 (demo: youtu.be/rcvSuBcBjyg).

Industry Experience

AI Integration Engineer | InfoAjax Consulting Oct 2025 - Nov 2025

Remote, Hyderabad, India

- Built a working proof of concept of a browser agent that reads screenshots, then clicks and types, using models on Azure AI Foundry. Also built a React, FastAPI and WebSocket console to watch agents live.

Software Engineer (Freelance, WSO2 project) | InfoAjax Consulting Oct 2024 - Jun 2025

Remote, Hyderabad, India

- Built REST APIs in WSO2 Micro Integrator that link the service-desk, billing and inventory systems of PLDT, a Philippine telecoms company. Supported them in production for four months under service-level agreements.

Technical Skills

Programming: Python, SQL, Java, C, Bash

NLP and ML: PyTorch, NumPy, Hugging Face Transformers, LoRA, scikit-learn, pandas; Transformer, CNN and BiLSTM encoders, character-level representations, Vision Transformers

Research practice: experiment design, pre-registration, bootstrap confidence intervals, data-leakage audits, public dataset release, IEEE manuscripts

Engineering: RunPod GPUs, Docker, Git, GitLab, Linux, FastAPI, REST, WebSocket, Azure AI Foundry

Awards and Leadership

1st Place, Global Challenge Lab, Imperial College London Jul 2026

- Led the AI and ML work for OrbitOps in a five-person team (£1,500 prize, over 1,000 students). OrbitOps is a reliability layer for AI on satellites that uses pre-launch fault injection, in-orbit known-answer probes and rollback.

Technical Lead, Student Activity Council, KL University 2023 - 2025

- Mentored over 20 junior students in machine learning and web development, and ran workshops for over 100 students. Received the SAC Momentum Award.